

EDCI 556 – Final Project Design Document

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Pitch

A game where players communicate to create tetrominos and solve tetris-style levels, to generate a bridge to save our little penguin friends, who are trying to cross a lava pit. Through its gameplay, players will be able to learn the spatial area and orientation that a tetromino occupies, and arrangements that will allow them to have the most complete set of tetrominos, laid out end-to-end. When played in pairs, players will also learn to effectively communicate to create the best tetromino, as needed.

This design document is an evolution of the initial pitch document, with each aspect explained thoroughly.

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Introduction/Overview

Blockrun is a PC game, comprising a mashup of three wildly popular genres: puzzle, adventure, and multiplayer. In the multiplayer mode, the game is a fast-paced two-player cooperative game where communication can save the day or kill cute penguins. Players must work together to build a bridge for the penguins to cross, using blocks made of ice. Blockrun has a level-based structure with a three-star tier scoring system – the more penguins saved, the more stars earned. Each level has an end goal for the penguins to reach

that progressively gets more difficult for both players. The difficulty increases by blocks being harder to create due to resources, special penguins, obstacles, and limited time to complete a level.

Target Audience and Implementation Context

The primary target audience for the game will be casual players and students in classrooms grades K-5. Players interested in learning to solve spatial puzzles in a limited time in a two-player setting, as well as students whose teachers are interested in facilitating cooperative engagement and teambuilding. As an educational game, it will be implemented in formal classrooms to teach collaboration, effective communication, clear instruction articulation and visuospatial reasoning.

Learning Objectives

- **Primary Objective:**
Students will analyze and evaluate blocks, shapes, and empty spaces in the game using visuospatial reasoning to demonstrate a comprehensive understanding of the shapes and manipulations of the shapes using rotations and translations.
- **Secondary Objective:**
To complete levels in the game, students will collaborate effectively in person as demonstrated verbally acknowledging other viewpoints and representing their own respectfully.

Hooks

Blockrun is engaging because each player (as an individual or a pair) will have to collect resources to fulfill objectives and must work together to build a bridge to get as many funny and cute penguins across a sea of scalding magma. At the end of each puzzle level, points will be awarded to players based on how well they did to get penguins to the goal. A single player mode will allow players to play levels individually, where the puzzle shapes are randomly generated, and a practice mode will be available for low-risk game play. In the multiplayer mode, two players must communicate and come up with the best shaped blocks, to form the bridge. It aims to be chaotic. It's puzzles, aimed to challenge. And of course, it has penguins.

Game Mechanics

There are two different game styles within the main game: adventurer and puzzle solver. The Adventurer has the following characteristics:

- All players start off as adventurers
- Ability to collect resources (block-building resources, coins, power ups, and environment interactions)
- They have a Third-Person point-of-view
- Ability to create tetrominos, from collected resources, based on instructions from puzzle solver
- Send tetrominos to puzzle solver (via environmental interaction)
- Review final shape created by the puzzle solver and provide feedback at the end of the level

The Puzzle Solvers have the following characteristics:

- An adventurer with a special role (in multiplayer mode, it is usually the one who hosts the game)
- Ability to pick up special resources called “keys” to open “puzzle caves”
- They have a First-Person point-of-view
- Puzzle solvers describe the type of tetrominos needed (based on attributes such as obstacles) to make the bridge
- Puzzle solvers utilize tetrominos to attempt to build a bridge for penguins to cross
- On puzzle completion, the puzzle solver assumes the role of a regular adventurer
- Review and rate created tetrominos on a scale and provide feedback at the end of the level

The puzzles follow traditional Tetris-styled controls with modifications:

- Rotation and translation of tetrominos
- Speed up drop-down time
- Filling up a row completely spawn the penguins on that row
- Once a piece reaches the end goal, then the penguins start crossing the bridge
- If there is no tetrominos “floor” beneath the penguin, they fall off the bridge, resulting in a loss of points. If all penguins fall off (regardless of the bridge’s completion status) the puzzle is declared a failure, and the player must start again.

Artificial Intelligence

This game contains non-playable characters (the penguins) that automatically move forward, once the bridge has been constructed. Penguins come in different types, which determines any special characteristics they have.

Other feedback providing mechanisms are doors, that can be opened with keys. When a key (or any collectable) is picked up, the HUD element updates that provides feedback to the player. When a door is interacted with, there are audio cues that play on the server (i.e. all connected clients can hear it), thus providing feedback.

There are level loaders present with the doors, that teleport player into the puzzle level. This acts as a trigger for the puzzle level, and after a short delay, the tetrominos start falling.

Game Elements

The game elements represent basic collectible “objects” and low poly visuals are employed for aesthetic. Sample game elements include block-creation resources, coins, jump-boost power ups, door keys, doors, respawn checkpoints.

Story Overview

“Two researchers have been dropped on a remote frozen wasteland, where they must collect resources and build bridges to save the penguins from extinction. Researcher One takes on the role of Adventurer collect block resources, which allows them to create shapes called tetrominos. The other researcher enters a puzzle cave and takes on the role of Puzzle Solver. They must communicate the layout of the floor, based on which the other researcher will create tetrominos and send it to them. The puzzle solver will then place the tetrominos to save as many penguins as possible.”

Game Progression

At the beginning of the session, the adventurer (or adventurers) will walk around collecting resources and looking for the key. Once the key is found, the puzzle solver will collect it, and then enter the locked entrance to the cave, to start the puzzle. If in the multiplayer mode, other player will stay in adventure mode, collect resources, and begin crafting blocks to send to the puzzle player. In this way, the puzzle player will use the crafted blocks as they arrive to build a bridge for the penguins to cross (in singleplayer mode, the blocks are randomized). The blocks will fall like Tetris with the exception that when a row is filled, instead of vanishing, it will freeze in place. Once the pieces hit the top of the stage, the level is over, and the penguins will move forward. If they have nothing to walk on, they will fall off the bridge. As the penguins cross the bridge, points will be awarded, and review will take place for both players before moving to the next level. In practice mode, players will start with puzzle mode to familiarize themselves with the controls, and adventure is not available.

User Interface

The color palette is based off colors found in snow-based biomes and general blue-white color scheme, simplistic font family (Arial), elements (buttons, sliders, checkboxes) follow low-poly 3D influence.

The user interface involves crucial functions, such as creating a lobby (to play the game in), character selection, changing the game settings, and disconnecting or exiting the game. There are also user interface elements that provide hints to players for the controls.

Implementation Artifacts and Guidelines

Provided with this game is a sample lesson plan for teachers to utilize in their classrooms (Appendix A). It is up to the discretion of the teacher how this game will be utilized in the classroom. The idea we envision is twofold: assisting students in developing visuospatial reasoning with 2-D shapes, rotations, and translations, and assisting students in developing collaboration and communication skills. Blockrun cannot provide this learning alone but can act as a tool or material to aid in the learning process.

Story Bible

This game does not have an official story beyond what is presented in the story overview.

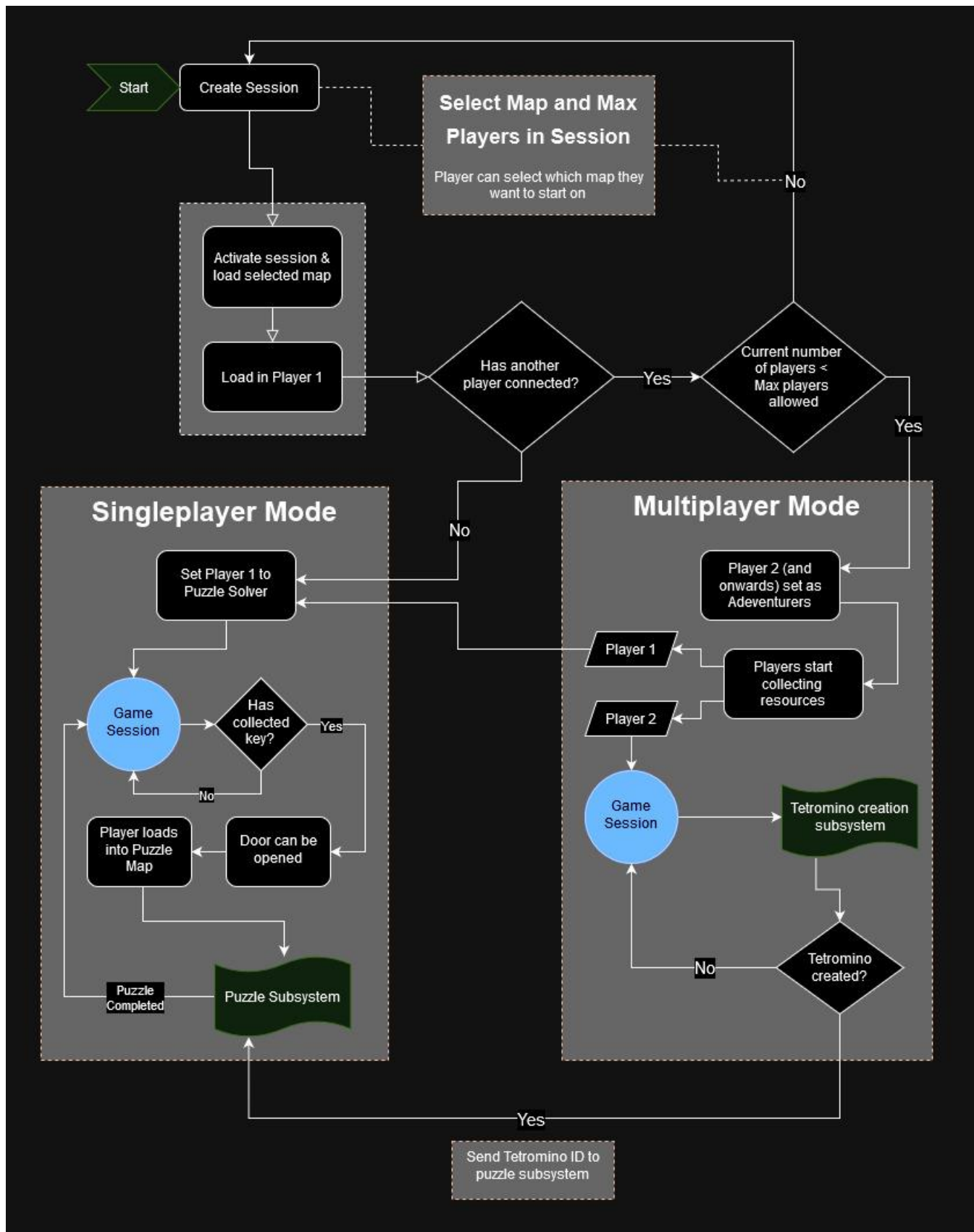
Script

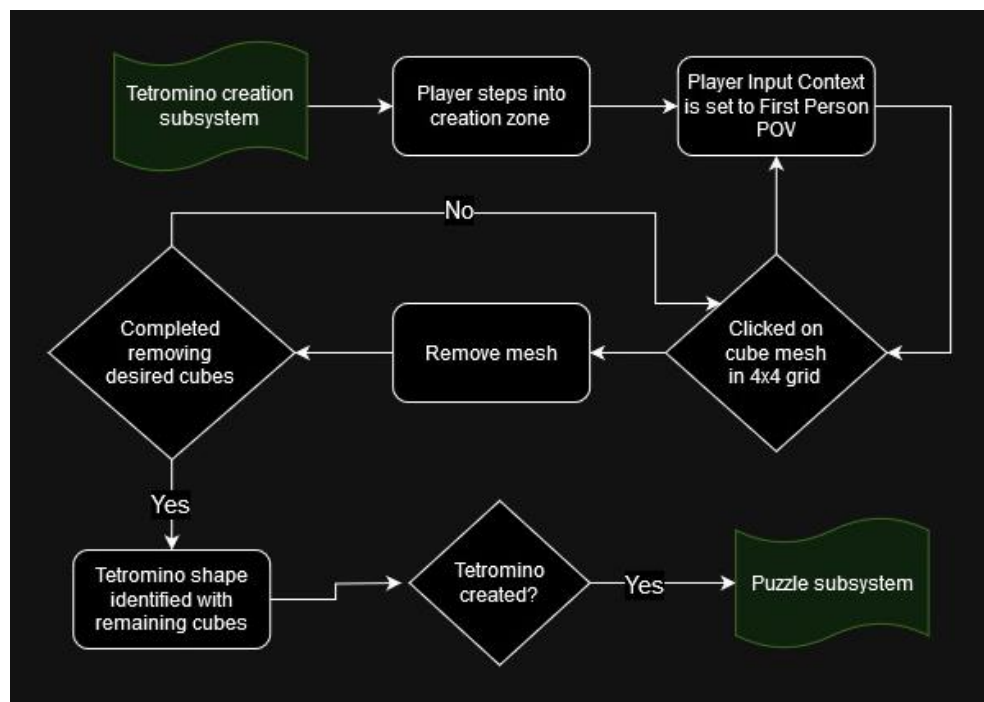
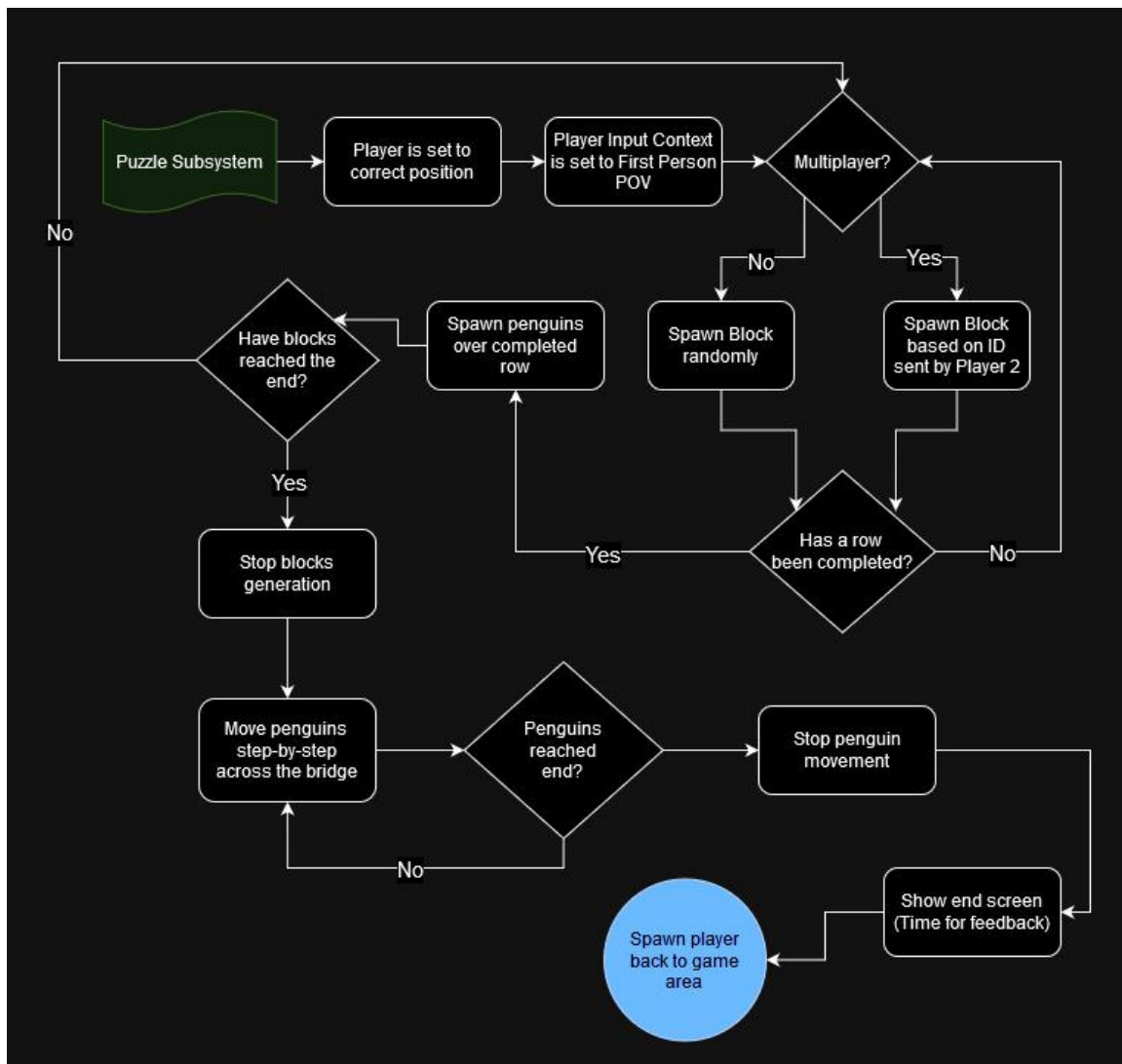
The only script is at the beginning of the game during loading which is the same as the story overview.

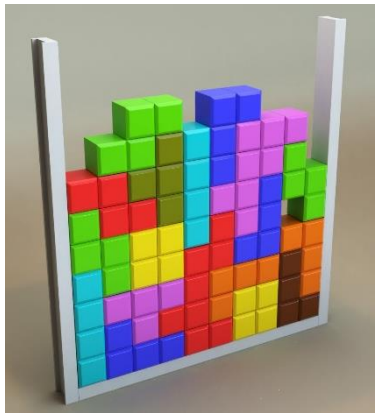
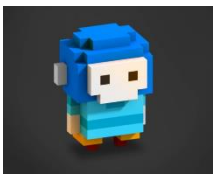
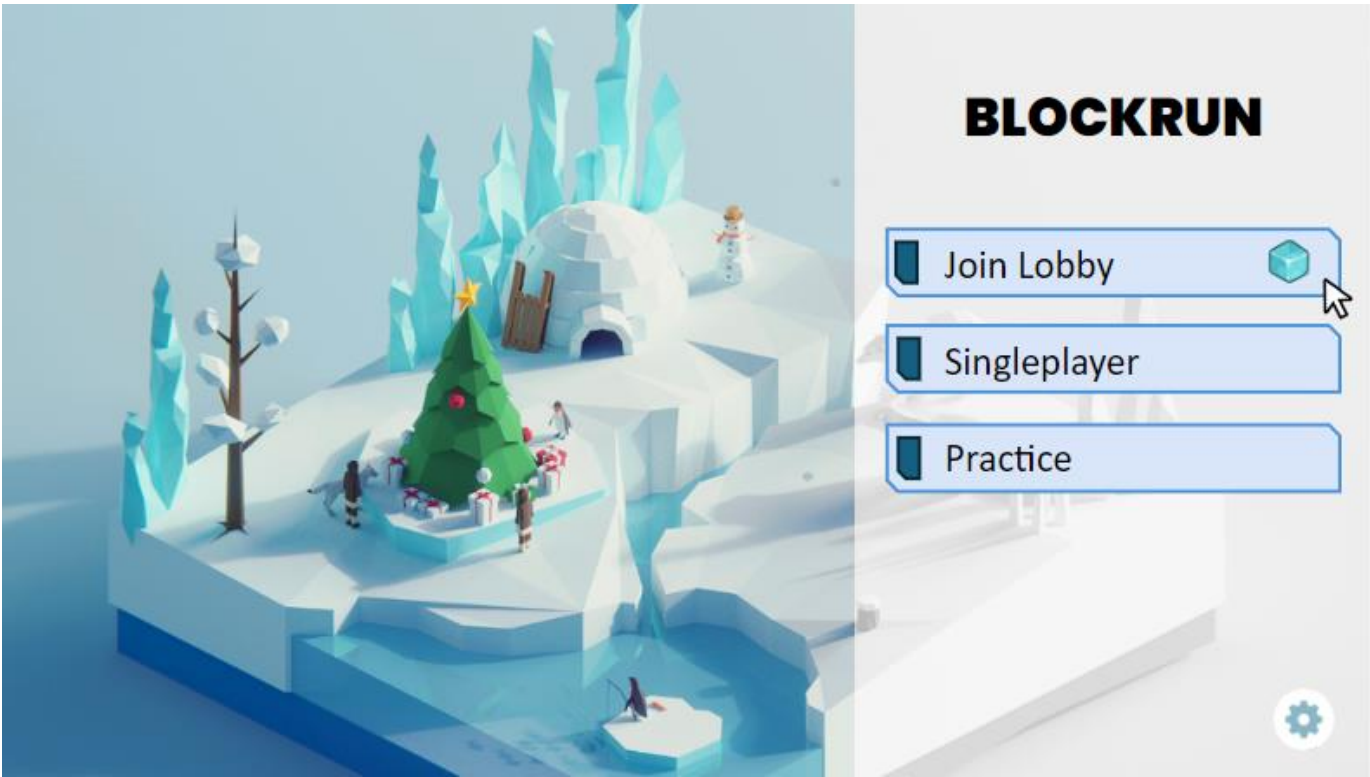
Game Minute

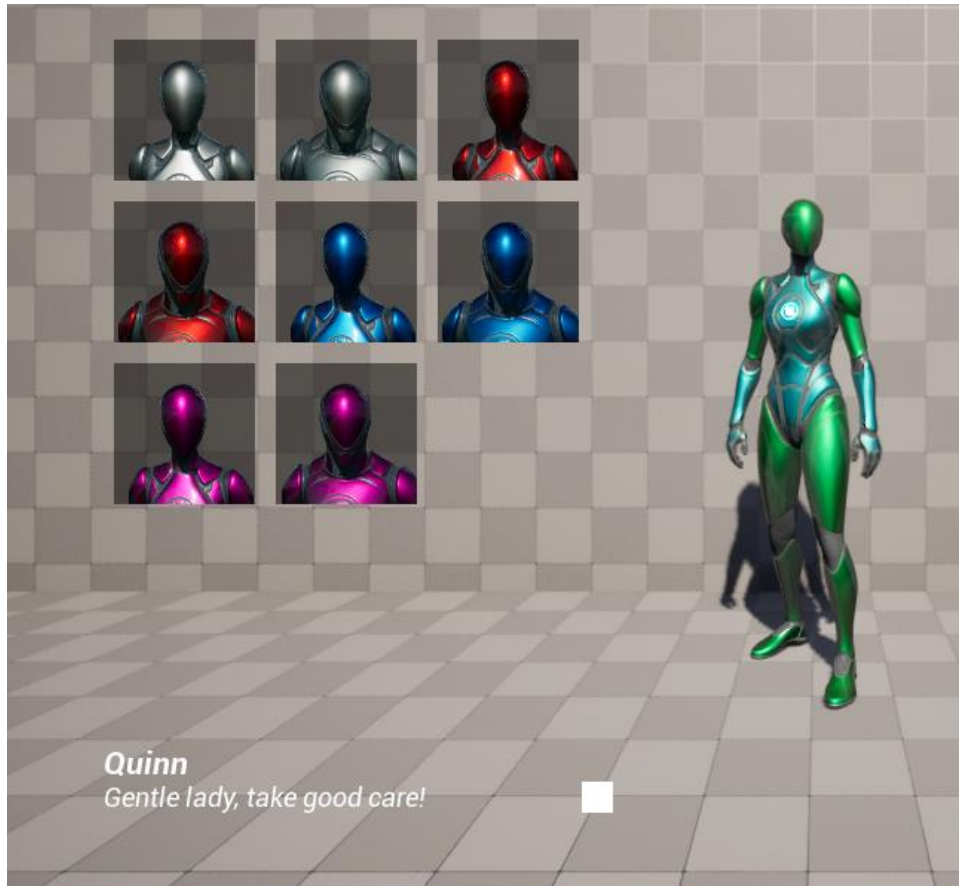
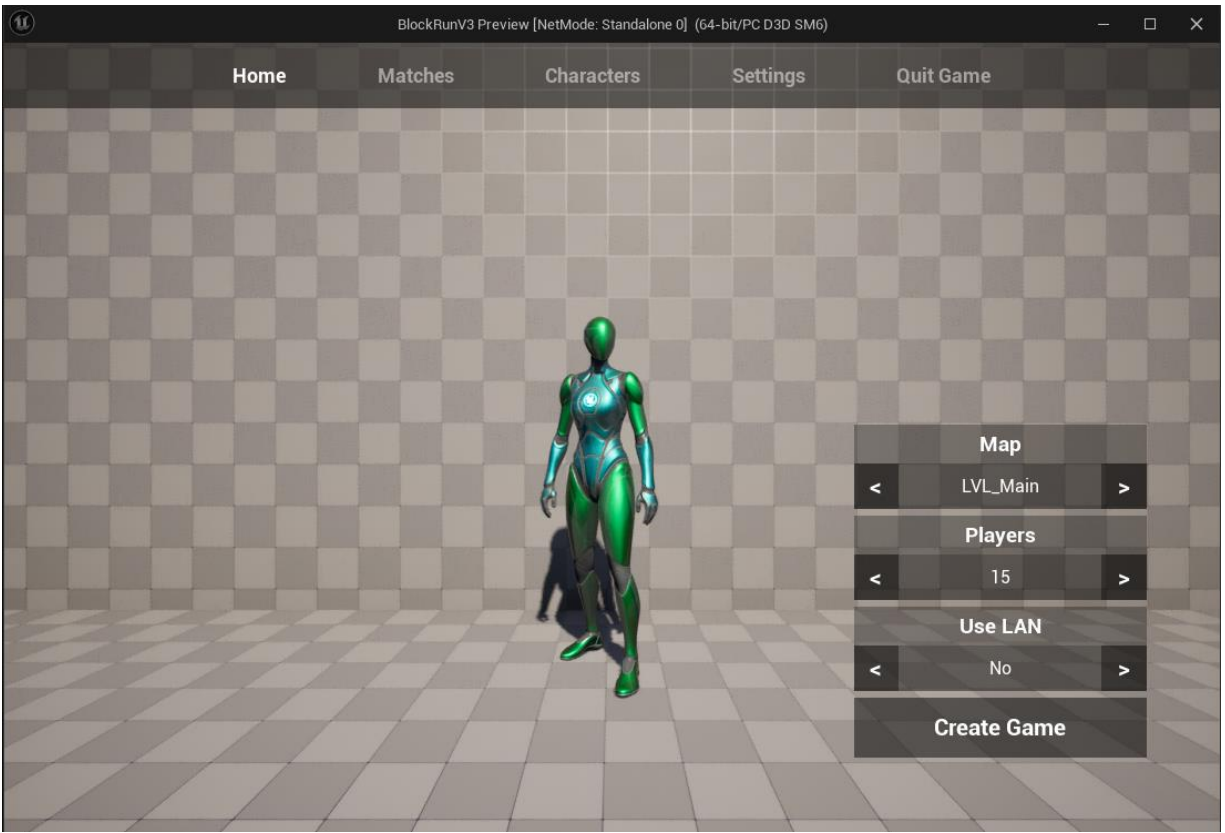
The game starts with the player (or players) in a world filled with resources, and a puzzle to unlock. A general gameplay minute will include the players collecting resources and a key to the puzzle cave door. At the beginning of a two-player session, both players will be in adventure mode walking around collecting resources and looking for the key. Once the key is found the player with the puzzle solver role can then enter the locked door and begin playing puzzle mode. The other player will stay in adventure mode and begin crafting blocks for the puzzle player to use. The puzzle player will use the crafted blocks as they arrive to build a bridge for the penguins to cross. The blocks will fall like Tetris with the exception that when a row is filled, instead of vanishing, it will freeze in place and the penguins will appear. Once the pieces hit the top of the stage, the penguins will start moving forward – if they have no floor to “walk” on, they will fall. Once all remaining penguins cross the bridge, the level is over, and the penguins will cross or fall. Points will be awarded based on the number of penguins saved, and a review will take place for both players before moving to the next level.

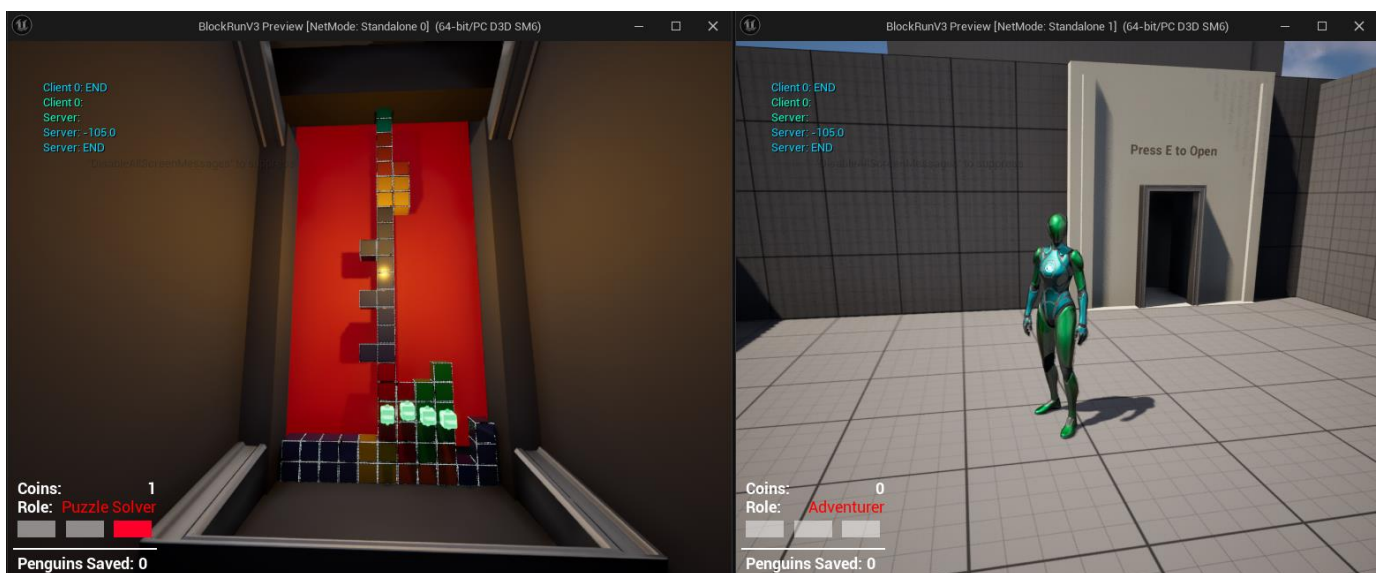
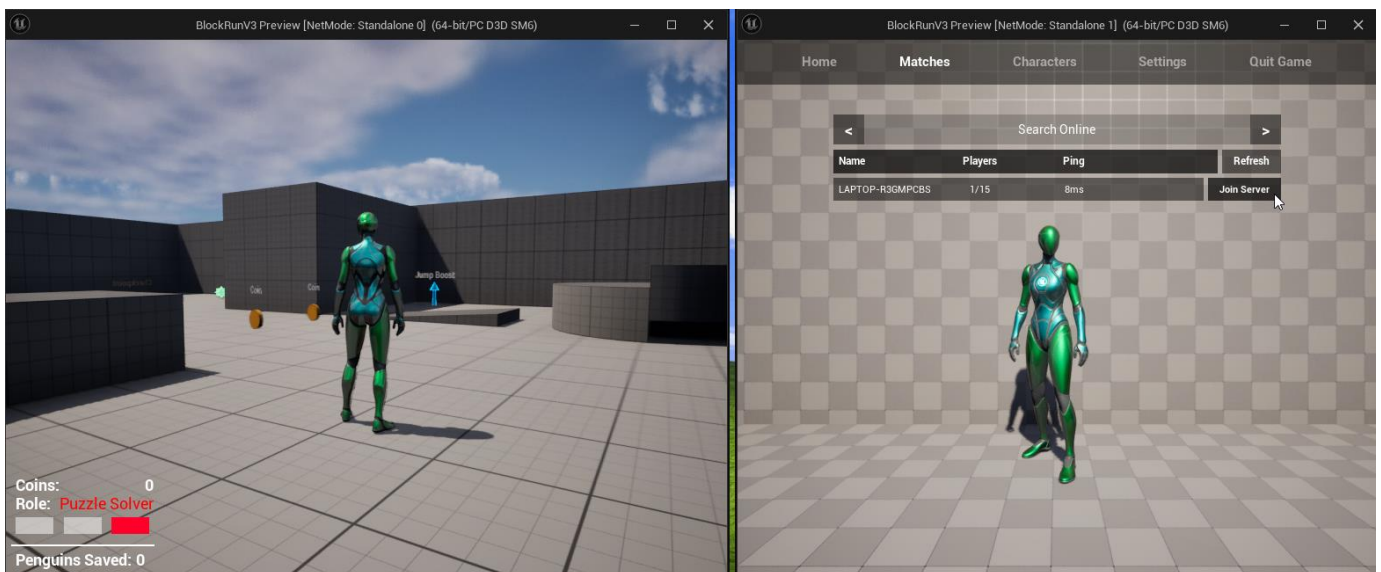
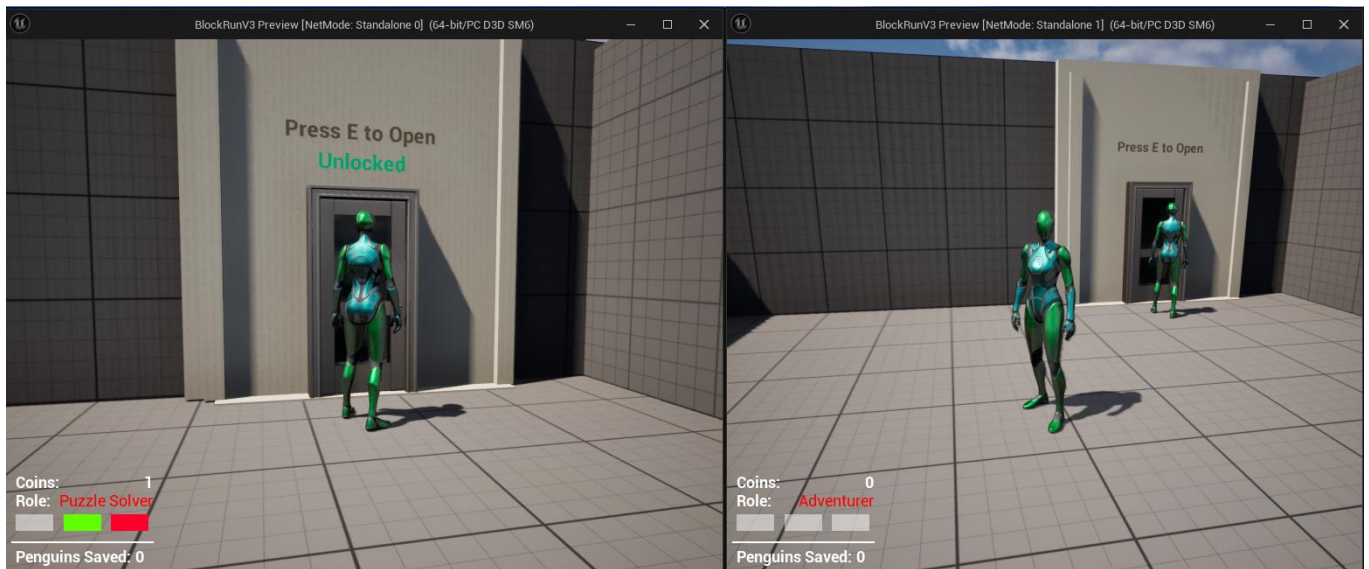
Flowcharts

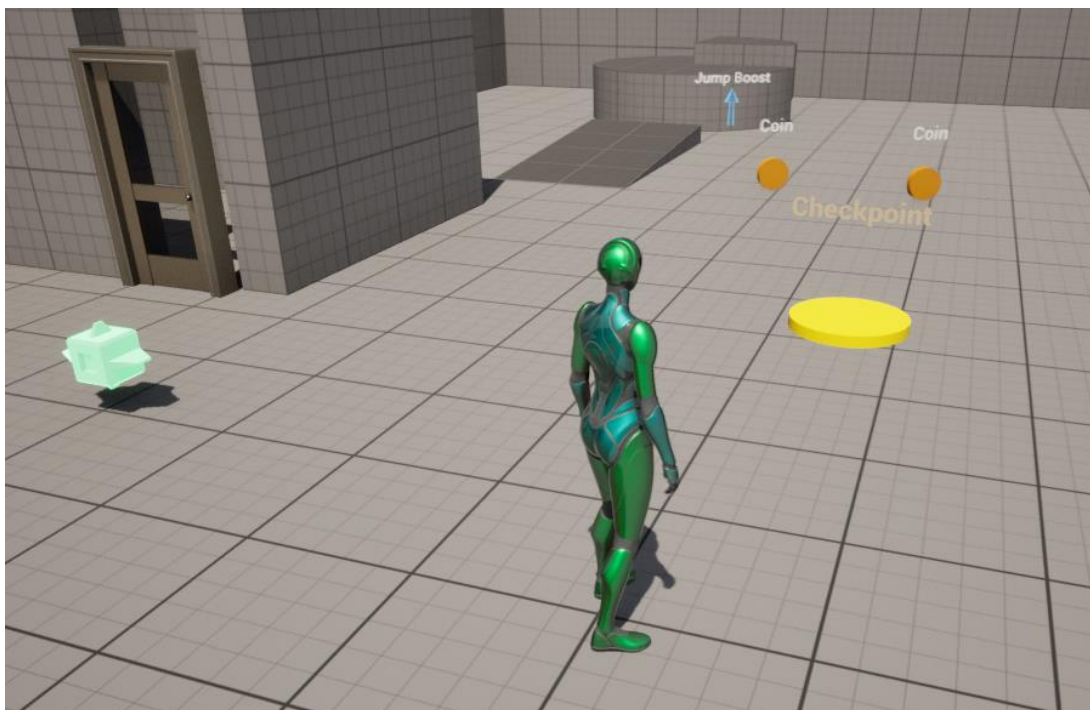
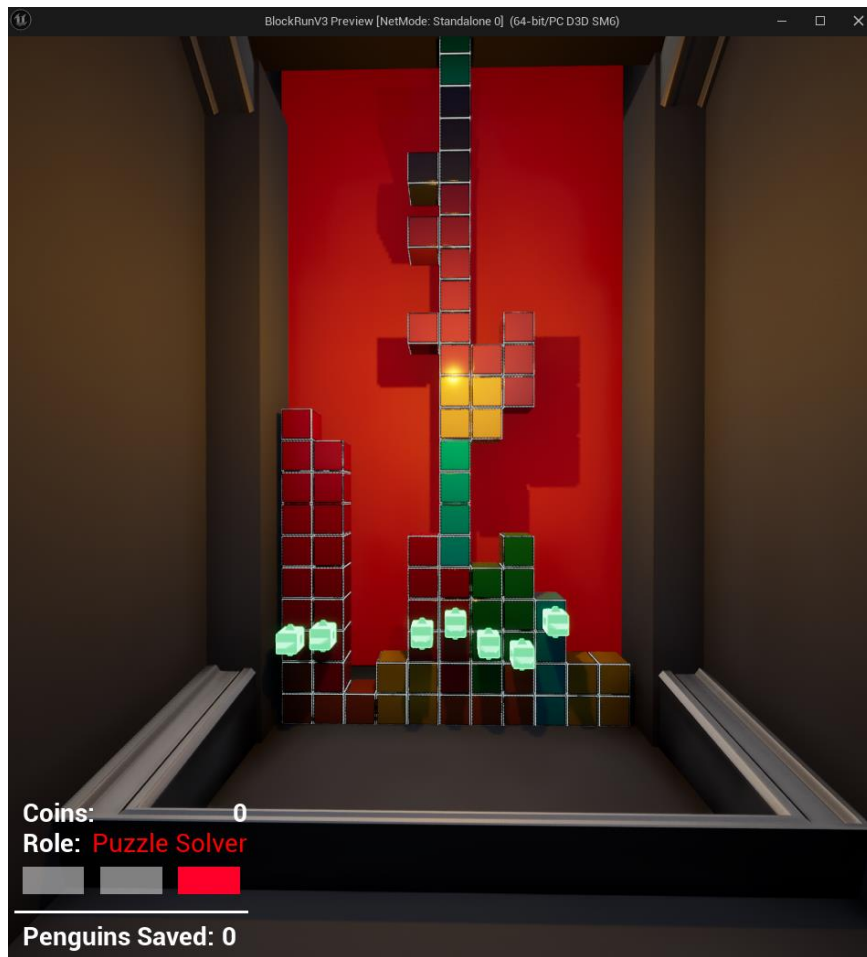












Technical Design Document

List of all features

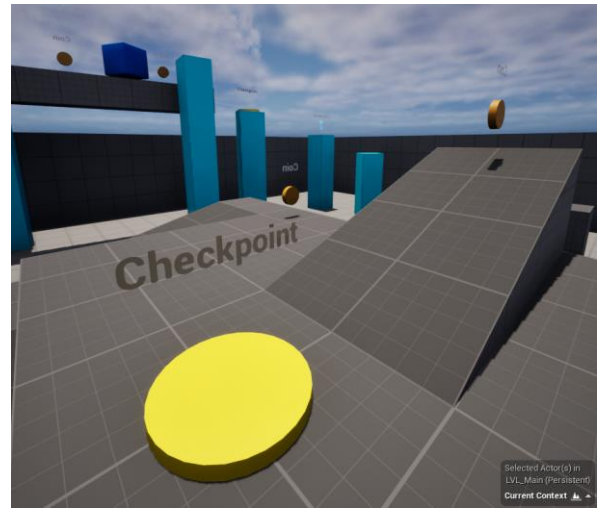
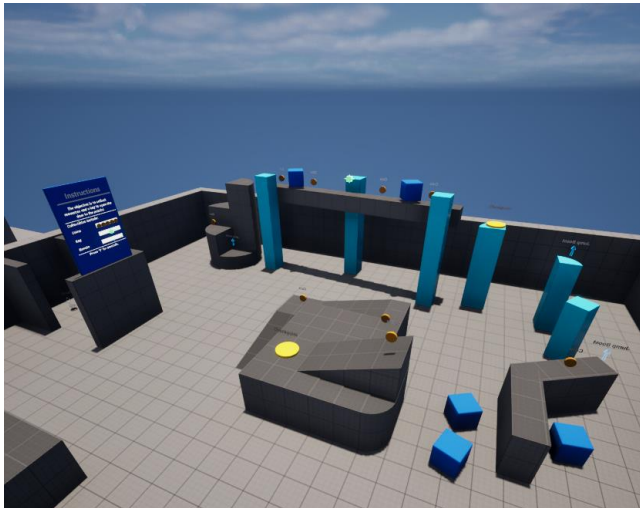
The features available in the game are character movement, character roles (adventurer and puzzle solver), character specific objectives, resource collection, power-ups collection (for traversal), lobby creation (for 2 player mode), practice maps, tetris-based puzzle solving, and tetromino generation (for use in the puzzle map).

Choice of Game Engine

The game engine used to develop the game is Unreal Engine 5 (5.3.2).

High Level Diagrams

All the levels follow the basic structure of a pair. One map will facilitate the adventurer, where collectibles will be spread around. The other map will facilitate the puzzle solver, where they use the collected key to open the puzzle cave and generate the bridge to save penguins.



Game Object Details

Each game object is an Unreal Engine blueprint, with atleast a static mesh component (for the visual shape), a box or capsule collider (to be able to interact with the Player Controller), and a toggle-able text render (that facilitates accessibility features of the game).

Physics Engine

The physics engine utilized is Unreal Engine's built in Physics Component system (Chaos Physics) to facilitate Rigidbody dynamics. This is used for in-scene actors' collision responses, raycasting for actor detection, and networked physics.

Game Logic and Artificial Intelligence

The game is built on a multiplayer facilitating systems logic. In the beginning, a server will create a session with a generic map. This will allow subsequent players (called clients) to join in as the second player in the map. The game has a central "CustomGameMode" class that acts as the server, and passes information to, from, and in-

between itself and the connected clients. The “CustomGameMode” is responsible for each connecting player to derive from a base “CustomPlayerController” that has two “InputContexts” built in. These input contexts allow the player to switch controls seamlessly between the adventurer mode and the puzzle solver mode. Player Controllers were used rather than player characters, since the multiplayer logic allows a Player Controller class to always exist on a server, when a session is online. This makes data tracking and variable tracking easier during runtime.

The server is responsible also for keeping track of which “CustomPlayerController” has collected which collectible and updates the heads-up display (HUD) of that specific player locally (i.e. only that client will see the changed values). The interactable objects (doors specifically) in the level detect the “CustomPlayerController” that possesses the key object. This is done through checks on the door locally, and once it opens, a server call is made to do the door open animation on all the connected clients. The door holds a level loader trigger box that then loads in the player to the puzzle solving map. Once loaded in, the server also changes the input context of the Player Controller and keeps track of the location to spawn the player back after finishing the puzzle.

This is where the game logic sees a fork as follows:

- If the singleplayer session is created, then the tetrominos inside the puzzle level are generated randomly. This allows the player to only focus on building the bridge and getting the highest points (which facilitates the primary goal of visuo-spatial learning).
- If the multiplayer (two player session) is created, then the other adventurer will collect resources and communicate to the puzzle solver. The puzzle solver will instruct on what shaped tetromino needs to be created, and the adventurer will have to do that in a special zone. The adventurer will be presented with a 4x4 grid of cubes. Looking and clicking on one will remove a cube (using raycasts) and then once the desired shape is achieved, a trigger is done to inform the “CustomGameMode” to detect the shape and assign it an ID based on one of the seven tetrominos. The “CustomGameMode” will send the data of the tetromino to the puzzle level, which will then spawn in that shape.

Audio and Visual Details and Specifications

The visuals in the game are targeted towards creating an atmosphere of being present in a snowy wasteland. There are glaciers, icescapes, frozen caves, frozen rivers, etc. that immerse the player in the atmosphere. The visuals are carried over into the user interfaces as well, with the same color palette. The character selection screen allows players to customize their character as well. The color palette can be seen below.

Environment Palette			
Character Palettes			

The audio in the game is simple. Collectables, on being picked up by the player, play a sound at the location (that is audible to any other nearby players). Interaction with world elements, such as doors opening, also has a sound played at its location which can be useful to all players connected to the server.

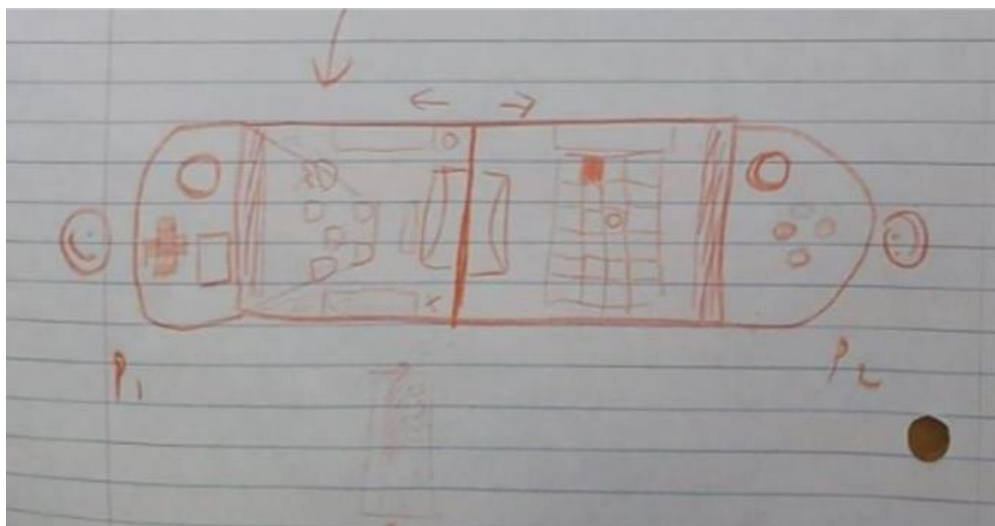
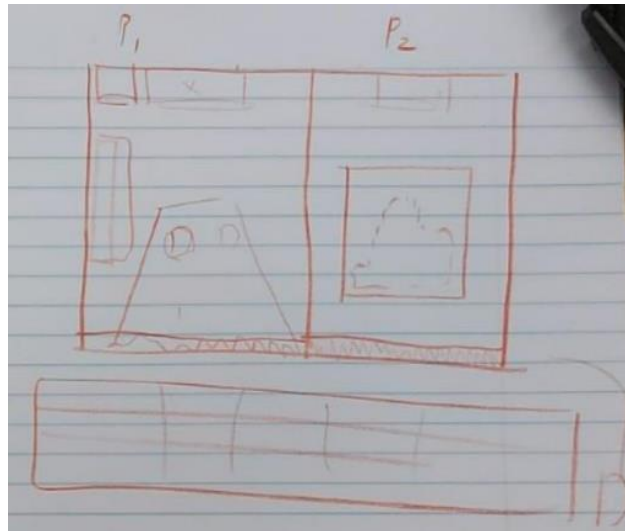
Networking

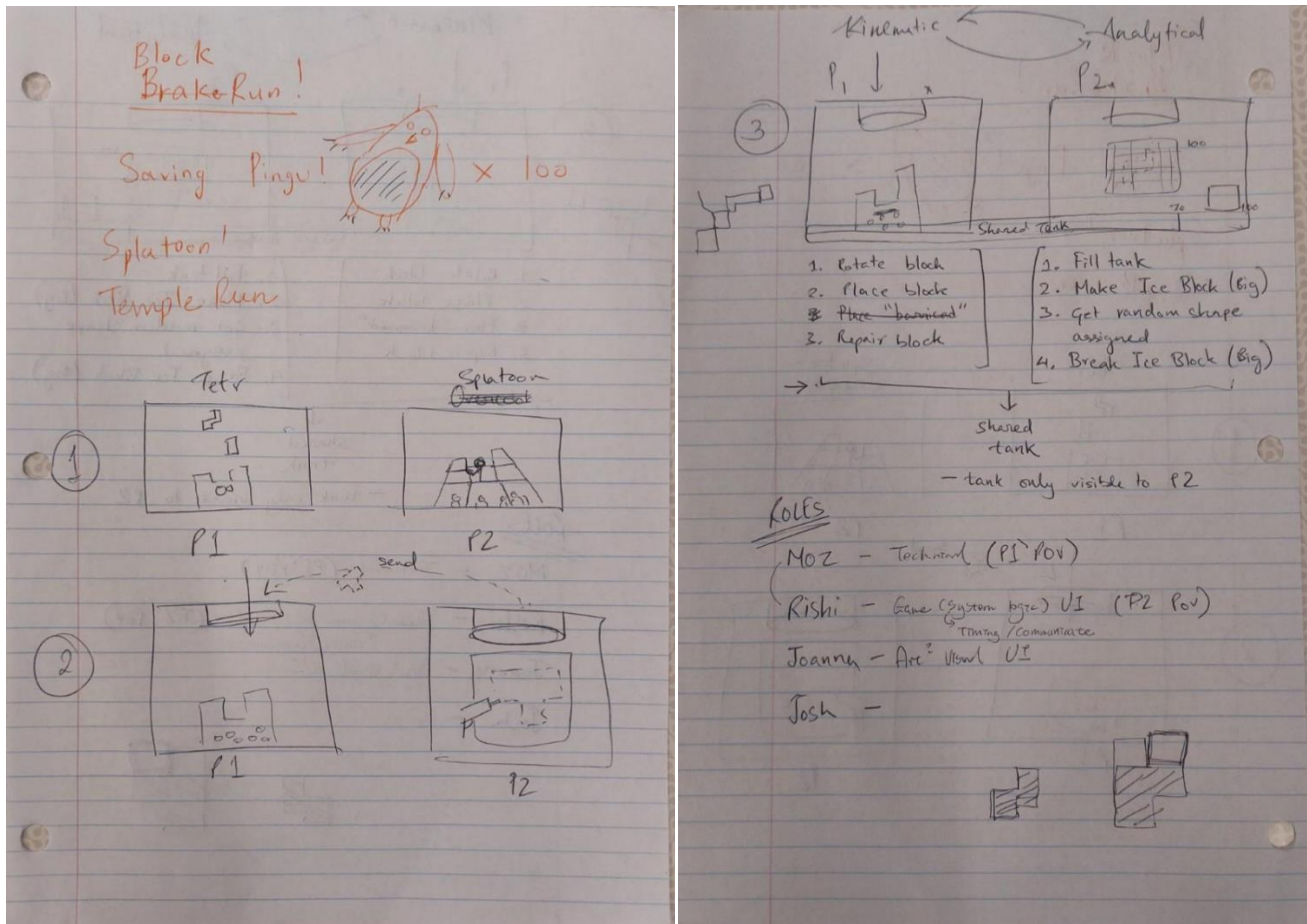
The game has single-player and multiplayer mode. In singleplayer mode, there is no communication between player characters. In multiplayer mode, the adventurers can affect the gameplay of the puzzle solvers by supplying them with construction blocks.

Delivery Platform and Hardware/Software Requirements

The game has been made to be played on a personal computer running Windows 10 and above. The minimum to run the game is Intel Core i5 (or equivalent), 8GB RAM, and 2GB of free storage space. The recommended specifications are Intel Core i7 (or equivalent), 16GB RAM and 10 GB of free storage space. The supported screen-sizes are the standard 16:9 and 16:10 aspect ratio screens (widescreen formats are not supported).

Storyboards





Schedules and Business Documents

The development of the game during the semester will be in three phases (and four parallel channels of responsibility for each member) iteratively:

	Channel A (Technical)	Channel B (Technical)	Channel C (Art & Design)	Channel D (Research)
Phase 1	Tetris-style component; base game development	Block-creation component; source control	UI, pre-production, storyboarding, creative direction	Research and development of learning outcomes
Phase 2	Multiplayer component; integration of Phase 1 elements	Block transfer component	Asset design, asset collection and management	Implementation and QC of learning outcomes
Phase 3	Integration of Phase 2 components; packaging of game for distribution	Finishing and testing	UI and UX integration	Game Testing, review, and feedback

Appendix A

Lesson Plan	Blockrun
Topic:	Visuospatial reasoning
Grade Level(s):	k-5
Objective	Students will analyze and evaluate blocks, shapes, and empty spaces in the game using visuospatial reasoning to demonstrate a comprehensive understanding of the shapes and manipulations of the shapes using rotations
Materials Needed:	- Paper - Pencil - Tablet or computer - Access to “BLOCK RUN” game
Sequencing of Lesson:	- Allow students to play game on their own or in pairs - After a few minutes, ask students to think about the movements they are making while playing. - After the game session is over, ask students to fill out attached “Reflecting on My Learning” worksheet. - Ask students to share and discuss as a class the various responses and strategies students came up with - Have students play again using the discussed ideas and strategies. - Repeat the play and reflection as needed.
Rationale and things to watch out for:	- Students may struggle with directions. For many this may be the first game or interaction with visuospatial reasoning and block positioning they have done and may need suggestions. - This lesson is very important because of its logical advancement. This lesson provides students with an opportunity to think critically, apply previous knowledge, as well as be creative and hands on. This lesson reaches kinesthetic, visual, and auditory learners.

Reflecting on My Learning

Name: _____

1. At the beginning of this game I could _____

2. What I learned during the game is _____

3. What I will try during the next game is _____

4. What I learned about shapes is _____
